Syllabus - Advanced Visualization and Presentation

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At its essence, the aim of data visualization is to move data and its meaning(s) and context(s) from some origin (spreadsheets, observed phenomena, etc.) to a larger audience. It's a spectrum of incredibly powerful tools for not just understanding and explaining facts, but also for shaping what those facts are and creating the narrative around them. By the end of this course, you will have thought through your role and responsibility in an evolving field, developed a set of best practices that is likely to continue to change, engaged with larger social currents toward your own goals, and strengthened your skills in R.

Because this is part of a professional studies program based on open source software and the ethos behind it, the course will be very hands-on and require everyone's willingness to contribute and participate. Instead of tests and graded homework assignments, we'll focus on practice, critique, and revision, building continuously on individual projects and shared tools.

Because data science and data visualization—and the software we use for both—change so quickly, a lot of the community's discourse happens in less formal settings, such as blogs, social media, podcasts, and workshops, rather than just traditional academic journals and books. Our readings will likewise fall along this spectrum, and you'll have some flexibility in what you read and share.

Above all, I want this to be a course that is useful to you as you build a career of critical engagement with data. The schedule has some flexibility so we can adjust things to fit your needs and interests. Please be willing to share what you want to learn, contribute resources, and ask for what you need of me and each other.

Objectives

The first half of the course will be focused on non-spatial data visualization; the second half will be focused on spatial data and how to integrate the two. Some of the principles we go over for non-spatial and spatial will differ, but objectives remain the same.

By the end of the course, students will:

- Have an understanding of the **basics of visual perception**, and how to use that knowledge to design data visualizations well
- Be familiar with the **grammar of graphics** framework to think about components and purposes of visual elements
- Be skilled in **programming in R** and using the ggplot2 data visualization ecosystem
- Know how to **give and receive constructive feedback** on visualizations, both their own and others', and to revise and improve upon their work
- Be able to **identify potential harms** done by inappropriate or misleading visualizations, and make corrections
- Be able to make, articulate, and argue for **good decisions** in designing charts and maps
- Have made many, many unpolished visualizations and several polished, presentation-ready ones

Successful students will finish the course with finished products for their portfolios of high enough quality to include with applications to jobs or other academic programs:

- 1–2 completed, **presentation-ready data visualization projects**
- **reproducible, documented code** that can be repurposed at another organization
- contributions to an **open source codebase**

Materials

Readings

All readings will be available to students for free. Many will be open source texts and have code available. Readings will be a mix of theory and practice.

The schedule of the course will roughly follow the structure of the book *Fundamentals of Data Visualization* (Wilke, 2019) Both the book and the source code used to write it are available for free online.

We'll also read portions of *R for Data Science* (Wickham et al., 2023) (also open source) and *How Charts Lie* (Cairo, 2019), as well as a variety of other sources of different media. I'll keep a running list of resources in the online class notes with other tutorials and references.

Software

This is a rough set of the software and tools we will use, with open source software in italics:

- *R programming language*
- *ggplot2 and related packages*
- *RStudio*, *Positron*, or another integrated development environment
- *Quarto,* a markdown-based publishing system from the same team as RStudio
- *git* for version control, GitHub for storage of version-controlled materials and submitting code
- Blackboard for assignments and announcements

Other tools

If at all possible, **you should have a laptop of your own** for this class. All the software we're using is free and open source, so you should be able to install everything on your computer. If you do not have a laptop, you can borrow one from the library, or, because we will be using git for version tracking and GitHub for storage, you can use a lab computer and make sure to upload your work regularly.

We'll be doing a lot of sketching by hand (you don't have to be good at drawing), so you'll need **a notebook and pens or pencils** that are nice to doodle with. I highly, *highly* recommend finding a graph paper or dotted notebook.

Schedule

The schedule has some flexibility built into it, but tentatively goes as follows:

Week	Section	Topic
1	Non-spatial data viz	Intro to data viz; writing good code
2		Encoding data to visuals; making good decisions
3		Tables; exploratory data visualization
4		Color, styleguides
5		Text & annotation; uncertainty
6		Audience & accessibility
7		Responsibility & ethics
8		Project 1 work session
9	Spatial data viz	Review of spatial viz & spatial encodings
10		Guiding readers
11		Building context
12		Manipulating geographies

Week	Section	Topic
13		Data viz for community action
14		Flex time / catch up
15		Project 2 work session

All assignments and due dates will be updated on Blackboard.

Class structure

To keep the class active, I'll aim to lecture for no more than 1 hour, leaving the rest of the time for labs and group discussion. We will have labs every session, except the last class before each project is due, when we'll spend the full class period working on projects.

Grading

In data visualization there aren't any perfectly right answers, and there aren't too many perfectly wrong ones either. As a result, rather than tedious quizzes and problem sets, your grade will reflect the effort you put into developing your process and your critical eye, and how successfully you create compelling stories with data.

Labs

For every lab, there will be 3 tasks that earn you points:

- Uploading your lab notebook (1-2 points based on effort, not correctness)
- Sharing a chart or map from your lab for feedback from your classmates (1 point)
- Providing *constructive* feedback to a classmate (1 point)

Submitting your lab

There will be one repo on GitHub where I post lab notebooks. You'll fork this repo (we'll walk through this together in the first session) so you have your own copy under your account. This also allows you to keep your repo up to date as I add each week's lab. As you work, you'll commit your changes and push them back to your repo. Whatever is there Tuesday before class is what I'll grade based on. You get 1 point for submitting something with a minimal amount of effort, and 2 points for having worked through all or most of the lab.

Feedback

At the end of each class, I'll remind everyone to share a chart on our discussion forum (TBD but probably Slack). (If you don't have anything yet you're comfortable sharing, you can do it throughout the week, but might miss out of getting feedback.) This does not have to be a finished chart; a work in progress or sketch is fine. If there are specific things you'd like feedback or advice on, feel free to let us know. You'll also need to provide at least one comment to a classmate.

Examples of constructive feedback:

- "I like the colors you chose. They grab my attention but are still easy to read since they're not too bright."
- "I don't quite understand the headline. Do you mean something like …?"
- "I think you need more contrast between the labels and the background. I'm having trouble reading this without my glasses. You might want to try…"

Examples of feedback that isn't constructive:

- "I hate these colors."
- "It's good."

Being able to give (and receive) feedback and critique rooted in kindness and respect is a crucial workplace skill that we'll practice. **Mean, rude, or unnecessarily harsh feedback will cost points.**

Projects

There will be 2 projects, one midterm and one final, that you'll be working on throughout the semester. Both will build upon the lab work, and you'll have time to work on them in class and receive feedback from myself and your peers. The first will be non-spatial data, and the second will be both spatial and non-spatial. You'll be responsible for moving from a dataset through to a polished visualization that tells a story and has real-world impact. You will also document your process along the way. Each project will also have a brief write-up to explain what you did and why, and to situate your work into the theory and principles we study.

Case studies

We'll have 2 case studies, again where one focuses on non-spatial and one on spatial data. For each, you'll choose 1 of 2 published visualization projects to study, as well as several readings from a list. You'll facilitate a sort of "discussion" between the authors of the readings you choose, the author of the visualization, and yourself, creating your own unique framework for understanding someone else's work.

Extensions

If you need an extension for an assignment, reach out *in advance* letting me know why you need the extension and when you think you can have the assignment done. Unless there's a serious disruption in your life, I'd prefer extensions to be a week at most. Since labs aren't graded, unless you aren't able to get a single thing done, you shouldn't need extensions for them.

Grading scale

Grades will be rounded to the nearest whole percent.

Grade	Percentage
A+	97% +
A	93-96%
A-	90-92%
B+	87-89%
B	83-86%
B-	80-82%
C+	77-79%
C	73-76%
C-	70-72%
D+	67-69%
D	63-66%
D-	60-62%
F	< 60%

Grade distribution

Category	Share of grade
Case studies	15%
Labs	30%
Project 1 visualization	20%
Project 1 write-up	5%

Category	Share of grade
Project 2 visualization	25%
Project 2 write-up	5%

Attendance

As grad students, your course load is one of many responsibilities you juggle, so I know things may come up that prevent you from getting to class. If you need to miss class or will be late, just let me know in advance (email or DM), and as long as absences don't become excessive, it should be fine. If there is some reason you'll need to miss class several times, such as chronic illness (after all, COVID's still here), just let me know and we can figure something out. If you can't attend class but are able to participate remotely, I can stream on Zoom or WebEx. If your absense is excused in advance and you cannot attend remotely, I'll give you a brief summary of what you missed. Beyond that it is up to you to catch up through the readings and course notes.

Each unexcused absence, except for a serious emergency (e.g. you got into a car accident on the way to campus), will cost you a point off your grade. Excused absences will not.

AI

I generally don't use AI (LLMs, code generators, etc) in my work. With the tools I've tried, I've mostly found that the code is low quality if it even runs at all. This is a class on creativity and critical thinking, and a proprietary computer program sold by a defense contractor cannot do those things for you. My goal is for you to become the sorts of developers and data visualizers that can build skills for yourselves, keep up with best practices, and weather industry changes.

That said, there are 2 areas where you will be allowed to use AI tools of your choosing:

1. Debugging code *you have written yourself*
2. Generating documentation that you then check and edit

If you go that route, **you're on your own** to choose, vet, pay for, and fight with your tools. Keep all prompts as comments in your code so you can trace back through your work later. Please don't ask me to debug AI-generated code for you (I won't). Please don't submit code with incorrect AI-generated documentation.

You are not allowed to use AI for written assignments. There is no formal writing in this class; there is very little writing at all, and it's all about your opinions and interests. I would rather read a case study handwritten on a napkin and full of misspellings than one that reads like a chatbot.

If I have serious suspicions that you have submitted AI-generated or heavily AI-assisted writing, I will run it through several AI checkers. If the majority flag it as likely AI, we'll have a conversation, and you'll have the opportunity to convince me it's your own work, or resubmit a handwritten assignment.

Let's just not go down this road. You're all adults, and I'd rather trust and treat you as such. Anything less isn't a good use of anyone's time, tuition money, or tokens.

Plagiarism and academic integrity

UMBC defines plagiarism as:

> The intentional or knowing representation of the words, ideas or work of others as one's own in an academic exercise. The appropriation of the language, ideas or thoughts of another and representation of them as one's own original work.

I take plagiarism seriously. If I find any evidence of plagiarism, we'll have a conversation to work out a solution, which, depending on the situation, can be anything from redoing the assignment with a penalty, to a formal disciplinary process.

Two examples of plagiarism that could arise in this course:

- Passing AI-generated writing off as your own (see above about AI use)
- Reusing text from readings without proper citation

If you have questions about this or need help, please just ask me before it becomes a problem. The Office of Academic Integrity also has resources to help understand how to avoid plagiarism.

UMBC policies and resources

Immigration and international students

UMBC's Office of International Students and Scholars puts out guidance and updates regarding changes in federal immigration policy, including details on entering the US. I try to keep track of these, but I encourage you all to keep track as well in order to support yourself and your peers.

From UMBC's general counsel:

> UMBC is not aware of any requests or actions concerning our campus from U.S. Immigration and Customs Enforcement (ICE) or other federal law enforcement agencies at this time. The information here is general guidance concerning requests from any external law enforcement agencies for information, records, or access to non-public areas of the campus.
>
> As a public institution, UMBC's campus is largely open to the public, but many spaces are restricted for reasons of privacy, safety, and operational needs—including residence halls, classrooms, laboratories, and administrative and faculty offices. To access such non-public areas, law enforcement must have a judicial warrant or subpoena. Additionally, federal privacy laws generally prohibit the release of information from a student's records, including to law enforcement, without a valid court order or subpoena.
>
> If you receive a request from external law enforcement for information, records, or access to non-public space on campus, **notify UMBC Police (410-455-5555)**, who will coordinate with relevant university officials and respond to the request. More specifically, we advise that you:
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> 1. Call UMBC Police, as noted above.
> 2. Inform the government/law enforcement agent that you are not authorized to provide access and that they should coordinate with UMBC Police.
> 3. Invite them to wait in a public area while you contact UMBC Police.
> 4. Do not accept a subpoena or warrant on behalf of the university yourself.
> 5. Do not physically block or interfere with enforcement action.
> 6. Document the interaction. You have the right to document the situation, including:
>   - Date and time
>   - Officer name and identification number
>   - Agency affiliation
>   - Details of the interaction
>
> **If you have further questions, please reach out to ogc@umbc.edu.**

Accessibility and Disability Accommodations, Guidance and Resources

From UMBC's Office of Equity and Civil Rights

Accommodations for students with disabilities are provided for all students with a qualified disability under the Americans with Disabilities Act (ADA & ADAAA) and Section 504 of the Rehabilitation Act who request and are eligible for accommodations. The Office of Disability Access and Resources (DAR) is the UMBC department designated to coordinate accommodations that creates equal access for students when barriers to participation exist in University courses, programs, or activities.

If you have a documented disability and need to request academic accommodations in your courses, please refer to the DAR website at accessibility.umbc.edu for registration information and office procedures.

DAR email: adasupport@umbc.edu

DAR phone: 410-455-2459

If you will be using DAR approved accommodations in this class, please contact the instructor to discuss implementation of the accommodations. During remote instruction requirements due to COVID, communication and flexibility will be essential for success.

Sexual Assault, Sexual Harassment, and Gender Based Violence and Discrimination

UMBC Policy in addition to federal and state law (to include Title IX) prohibits discrimination and harassment on the basis of sex, sexual orientation, and gender identity in University programs and activities. Any student who is impacted by sexual harassment, sexual assault, domestic violence, dating violence, stalking, sexual exploitation, gender discrimination, pregnancy discrimination, gender-based harassment, or related retaliation should contact the University's Title IX Coordinator to make a report and/or access support and resources. The Title IX Coordinator can be reached at ecr@umbc.edu or 410-455-1717.

You can access support and resources even if you do not want to take any further action. You will not be forced to file a formal complaint or police report. Please be aware that the University may take action on its own if essential to protect the safety of the community.

If you are interested in making a report, please use the Online Reporting/Referral Form. Please note that, if you report anonymously, the University's ability to respond will be limited.

Notice that Faculty and Teaching Assistants are Mandated Reporters with Mandatory Reporting Obligations

All faculty members and teaching assistants are considered Mandated Reporters, per UMBC's Interim Policy on Sex Discrimination, Sex-Based Harassment, and Sexual Misconduct. Faculty and teaching assistants therefore required to report all known information regarding alleged conduct that may be a violation of the Policy to the Title IX Coordinator, even if a student discloses an experience that occurred before attending UMBC and/or an incident that only involves people not affiliated with UMBC. Reports are required regardless of the amount of detail provided and even in instances where support has already been offered or received.

While faculty members want to encourage you to share information related to your life experiences through discussion and written work, students should understand that faculty are required to report past and present sexual harassment, sexual assault, domestic and dating violence, stalking, and gender discrimination that is shared with them to the Title IX Coordinator so that the University can inform students of their rights, resources, and support. While you are encouraged to do so, you are not obligated to respond to outreach conducted as a result of a report to the Title IX Coordinator.

If you need to speak with someone in confidence, who does not have an obligation to report to the Title IX Coordinator, UMBC has a number of Confidential Resources available to support you:

Retriever Integrated Health (Main Campus): 410-455-2472; Monday – Friday 8:30 a.m. – 5 p.m.; For After-Hours Support, Call 988.

Center for Counseling and Well-Being (Shady Grove Campus): 301-738-6273; Monday-Thursday 10:00a.m. – 7:00 p.m. and Friday 10:00 a.m. – 2:00 p.m. (virtual) Online Appointment Request Form

Pastoral Counseling via The Gathering Space for Spiritual Well-Being: 410-455-6795; i3b@umbc.edu; Monday – Friday 8:00 a.m. – 10:00 p.m.

Women's, Gender, and Equity Center (open to students of all genders): 410-455-2714; womenscenter@umbc.edu; Monday – Thursday 9:30 a.m. – 5:00 p.m. and Friday 10:00 a.m. – 4 p.m.

Other Resources

Shady Grove Student Resources, Maryland Resources, National Resources.

Child Abuse and Neglect

Please note that Maryland law and UMBC policy require that faculty report all disclosures or suspicions of child abuse or neglect to the Department of Social Services and_/_or the police even if the person who experienced the abuse or neglect is now over 18.

Pregnant and Parenting Students

UMBC's Interim Policy on Sex Discrimination, Sex-Based Harassment, and Sexual Misconduct expressly prohibits all forms of discrimination and harassment on the basis of sex, including pregnancy. Resources for pregnant, parenting and breastfeeding students are available through the University's Office of Equity and Civil Rights. Pregnant and parenting students are encouraged to contact the Title IX Coordinator to discuss plans and ensure ongoing access to their academic program with respect to a leave of absence – returning following leave, or any other accommodation that may be needed related to pregnancy, childbirth, adoption, breastfeeding, and/or the early months of parenting.

In addition, students who are pregnant and have an impairment related to their pregnancy that qualifies as disability under the ADA may be entitled to accommodations through the Office of Disability Access and Resources (DAR).

Religious Observances & Accommodations

UMBC Policy provides that students should not be penalized because of observances of their religious beliefs, and that students shall be given an opportunity, whenever feasible, to make up within a reasonable time any academic assignment that is missed due to individual participation in religious observances. It is the responsibility of the student to inform the instructor of any intended absences or requested modifications for religious observances in advance, and as early as possible. For questions or guidance regarding religious observances and accommodations, please contact the Office of Equity and Civil Rights at ecr@umbc.edu.

Hate, Bias, Discrimination and Harassment

UMBC values safety, cultural and ethnic diversity, social responsibility, lifelong learning, equity, and civic engagement.

Consistent with these principles, UMBC Policy prohibits discrimination and harassment in its educational programs and activities or with respect to employment terms and conditions based on race, creed, color, religion, sex, gender, pregnancy, ancestry, age, gender identity or expression, national origin, veterans status, marital status, sexual orientation, physical or mental disability, or genetic information.

Students (and faculty and staff) who experience discrimination, harassment, hate, or bias based upon a protected status or who have such matters reported to them should use the online reporting/referral form to report discrimination, hate, or bias incidents. You may report incidents that happen to you anonymously. Please note that, if you report anonymously, the University's ability to respond may be limited.

UMBC Writing Center

From the Academic Success Center:

The Academic Success Center offers free writing assistance through our Writing Center, which is located on the first floor of the Library. We also offer online and asynchronous tutoring. Writing tutors are students like you who receive ongoing training to stay up-to-date on the best tutoring techniques. To make an appointment, please visit http://academicsuccess.umbc.edu/writing-center

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Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for data science* (2nd ed.). O'Reilly Media, Incorporated. https://r4ds.hadley.nz/

Wilke, C. O. (2019). *Fundamentals of Data Visualization*. https://clauswilke.com/dataviz/